

SCQ (Single Correct Type) :

- Chromite ores + $\text{N}_2\text{CO}_3 \xrightarrow{\Delta/\text{air}} \text{A} + \text{B}(\downarrow) + \text{C}(\text{gas})$

[$\therefore$ A is soluble in H_2O]

"A" + conc. $\text{H}_2\text{SO}_4 \longrightarrow$ "D" +

"D" + $\text{KC}_1 \longrightarrow$ Orange colour solid (E)

The correct statement about "E" is :

(A) M-O-M has bond angle is 126°

(B) "E" has two tetrahedral unit sharing with one corner of oxygen

(C) The bond length of bridging oxygen is longer than same terminal metal oxygen

(D) All of these
- An alloy of lead (valency = 2) and thallium (valency = 1) containing 65% Pb and 35% Tℓ, by weight, can be electroplated onto a cathode from a solution. How many gram of this alloy will deposit in 9.65 hours using a constant current of 0.5 amp?

[Given that only these two elements deposit simultaneously in given mass ratio at cathode]

[Atomic weight Pb = 208, Tℓ = 200]

(A) 3.7 gm (B) 22.5 gm (C) 37.1 gm (D) 24.9 gm
- For the spinel structure (MgAl_2O_4), the **incorrect** statement is

(A) 50% OVs are occupied by ions (B) Al^{3+} is equally distributed in TVs and OVs

(C) Oxide ions occupy ccp lattice (D) 12.5% TVs are occupied by ions
- Two flasks A and B have equal volume Flask A is maintained at 300 K & flask B is maintained at 600 K. Flask A contains $\text{F}_2(\text{g})$ and Flask B Ne. Assuming ideal behaviour of both gases and Collision diameter to be equal, select the correct option if masses of both the gases are same.

(A) Collisions made by F_2 molecules per second will be more than collisions made by Ne molecules per second.

(B) Each Ne molecule will move faster as compared to each F_2 molecule.

(C) Average of magnitude of momentum change due to wall collisions will be more in case of Ne as compared to F_2 .

(D) Ne molecules on an average will travel longer distances between successive collisions as compared to F_2 .

5. Choose the incorrect statement.
- (A) If the isothermal reversible expansion stage is replaced by an irreversible isothermal expansion in the Carnot cycle, cyclic integral of $\frac{dQ_{rev}}{T}$ will be negative.
- (B) Two reversible adiabatic paths for a gas can never cross.
- (C) All adiabats on PV curve are isoentrops.
- (D) Entropy change of an isolated system is always positive.

MCQ (Multiple Correct Type) :

6. $K_4[Fe(CN)_6] + \text{conc. } H_2SO_4 \longrightarrow P(g)$
 $KMnO_4 + \text{conc. } H_2SO_4 \longrightarrow Q(s)$ (Containing Mn)
 Correct statement regarding P and Q are:
- (A) P is CO_2 & Q is explosive substance
- (B) P is neutral gas & Q is MnO_2
- (C) Formation of P involved non redox change & Q is Mn_2O_7
- (D) P is neutral gas & Q is explosive substance
7. Analysis of distribution of molecular speeds of an ideal gas can be mathematically represented by the equation

$$\frac{\Delta N}{N} = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} e^{-Mu^2/2RT} u^2 du$$

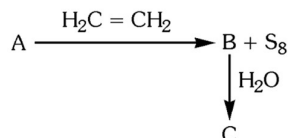
where $\frac{\Delta N}{N}$ is fraction of molecule having speed between u to u + du **M** molecular mass, **T** is

absolute temperature. The graph of 'fraction of molecules' vs 'speed' when plotted was found to be 'unsymmetrical' with very few molecules with very high or low speed and mostly molecules possessing speeds in the range of **270–330 m/s**. These graphs are termed as speed distribution curves. On the basis of the above information mark the statements which is/ are correct.

- (A) If absolute temperature of O_2 is half to that of SO_2 gas than the speed distribution curves will be exactly same.
- (B) The speed possessed by most of the molecules and average speed has the same value.
- (C) At same temperature and at lower speed, fraction of molecules of SO_2 will be greater.
- (D) At higher speed the fraction of molecules increases with increasing speed.
8. Ammonia gas evolved completely from 2.5 gm of a complex $[M(NH_3)_x(H_2O)Cl]Cl$ was passed through 60 ml of 1 M H_2SO_4 solution. After the reaction was over, excess acid required 140 ml of 0.5 M NaOH solution. **[Given : Atomic mass of M is 43]**
 Select the correct option(s).
- (A) The moles of H_2SO_4 reacted with given NaOH solution is 0.07
- (B) The mass % of ammonia in the complex compound is 60
- (C) The value of x in the formula is 4
- (D) 12.5 ml of 1M $AgNO_3$ solution is required for complete precipitation of Cl^- from the given compound

9. Which of the following is/are correct statement(s) ?
- (A) Probability of finding the electron in bonding molecular orbital is more than combining atomic orbitals
- (B) Bonding molecular orbitals are formed when same sign of orbitals will overlap
- (C) d-d combination of atomic orbitals gives δ and δ^* molecular orbitals.
- (D) at least one nodal plane will exist in the formation of ABMO.

10. $S_8 + 4Cl_2 \rightarrow 4A$ (Balanced chemical reaction)

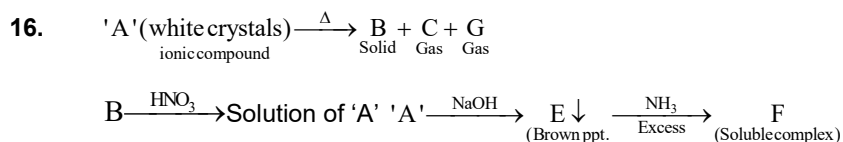


Then:

- (A) $A = SCl_2$
- (B) $B = Cl-CH_2-CH_2-S-CH_2-CH_2-Cl$
- (C) Reaction B to C is very fast
- (D) Reaction B to C goes slowly
11. 80.0 gm salt of weak base & strong acid XY is dissolved in water and formed 2 litre of aqueous solution. The pH of the resultant solution was found to be 5 at 298 K. If XY forms CsCl type crystal having r_{X^+} (radius of X^+) = 1.6 Å & r_{Y^-} (radius of Y^-) = 1.864 Å then select correct statement(s).
(Given: $K_o(XOH) = 4 \times 10^{-5}$; $N_A = 6 \times 10^{23}$)
- (A) Molar mass of salt is 100 g/mol
- (B) % Degree of dissociation of salt is 0.25
- (C) Edge length of AB is 4Å
- (D) Density of solid salt XY is 2 in gm/cc
12. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult's law. The normal boiling point of pure acetone is less than that of pure carbon disulphide. Select the incorrect statement(s) among the following.
- (A) Boiling point of mixture is always less than the boiling point of acetone.
- (B) Boiling point of the azeotropic mixture is always less than the boiling point of pure carbon disulphide.
- (C) When a small amount of carbon disulphide is added to an excess of acetone, boiling point of solution increases.
- (D) A mixture of acetone and carbon disulphide can be completely separated by simple fractional distillation.
13. $4AgNO_3 + 2H_2O + H_3PO_2 \rightarrow 4Ag + 'X' + 'Y'$
- If X is oxoacid of nitrogen and Y is oxoacid of phosphorous then correct statements is/are
- (A) 'X' is HNO_3
- (B) 'Y' is H_3PO_4
- (C) H_3PO_2 act as good reducing agent
- (D) 'Y' is H_3PO_3

14. Which of the following statements is/are **correct**?
- (A) Nitrogen (IV) oxide in basic aqueous solution disproportionates
- (B) Copper (II) nitrate and lead (II) nitrate each on thermal decomposition liberate both NO_2 and O_2 gases
- (C) Some metals (Cr, Al etc.) do not dissolve in cone. HNO_3 due to the formation of a passive film of oxide on the surface
- (D) Hydrogen peroxide shows reductive bleaching.

15. Select the incorrect statements for $[\text{M}(\text{AB})_2\text{b}_2\text{cd}]$
- (A) All geometrical isomers are optically active.
- (B) It has four trans isomer with respect to 'b'.
- (C) It has seven geometrical isomers.
- (D) It has four cis and two trans isomers with respect to 'b'.



Select correct statement(s) for species A to H

- (A) Both gases are elemental forms.
- (B) Solution of 'A' gives white ppt. with Na_2CO_3 solution.
- (C) F has coordination number of metal = 4.
- (D) ppt. E is soluble in excess KCN solution.
17. Match the precipitates of the compounds listed in List-I with the solvent(s) listed in List-II.

List-I

- (P) $\text{Zn}(\text{OH})_2$ precipitate dissolves in
- (Q) $\text{Cr}(\text{OH})_3$ precipitate dissolves in
- (R) AgCl precipitate dissolves in
- (S) CuS precipitate dissolves in

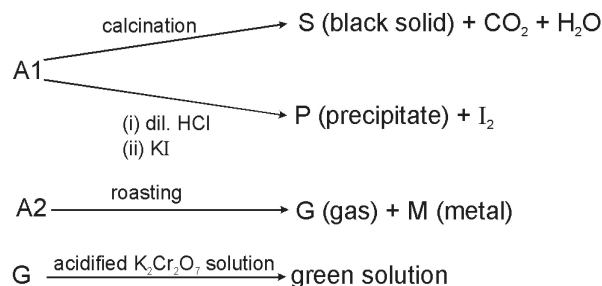
List-II

- (1) Potassium cyanide
- (2) Ammonia solution
- (3) Sodium hydroxide
- (4) Hydrochloric acid

The correct possible matching can be:

- (A) $\text{P} \rightarrow 3, \text{Q} \rightarrow 2, \text{R} \rightarrow 1, \text{S} \rightarrow 1$
- (B) $\text{P} \rightarrow 2, \text{Q} \rightarrow 3, \text{R} \rightarrow 3, \text{S} \rightarrow 2$
- (C) $\text{P} \rightarrow 3, \text{Q} \rightarrow 4, \text{R} \rightarrow 2, \text{S} \rightarrow 1$
- (D) $\text{P} \rightarrow 4, \text{Q} \rightarrow 3, \text{R} \rightarrow 1, \text{S} \rightarrow 4$

18. Two ores A1 and A2 of Copper show the following reactivity:



Select the correct options.

- (A) A1 = CuCO_3 anhydrous (B) A2 = Cu_2S
 (C) P = Cu_2I_2 (D) G = Can act as oxidising as well as reducing agent
19. The density of 2m aqueous H_2O_2 solution is 1.068 gm/ml. Which of the following is/are correct concentration(s) of this solution, in different units ?
- (A) Molarity = 2M (B) Volume strength = '22.4 V'
 (C) % (w/v) = 6.8 (D) Mole-fraction of water = $\frac{250}{259}$

Match the column

20. For an endothermic reversible reaction $4\text{A(g)} + \text{B}_2\text{(g)} \rightleftharpoons 2\text{A}_2\text{B(g)}$

Neglect the slope of change in concentration when system approaches equilibrium. Assume that changes are carried out very fast.

Column-I		Column-II (changes is)	
(P)		(1)	Increase in temperature
(Q)		(2)	Increase in pressure
(R)		(3)	Addition of A_2B at equilibrium

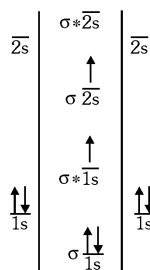
(S)		(4)	Addition of inert gas at constant pressure
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Codes:

	P	Q	R	S
(A)	(1)	(2)	(3)	(4)
(B)	(4)	(3)	(2)	(1)
(C)	(3)	(1)	(4)	(2)
(D)	(2)	(4)	(3)	(1)

(Numerical Value type) :

21. The MO electronic configuration of X_2 can be represented as follows



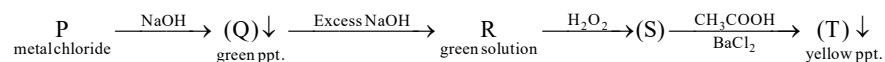
Following conclusion can be drawn:

- (i) It is excited state electronic configuration of X_2
- (ii) It is more stable than the ground state of X_2 molecule
- (iii) Bond order of X_2 in excited state is one
- (iv) It is more likely to dissociate into two X-atoms in ground state than in excited state

How many of the above conclusions are correct from given MO diagrams ?

22. If all compounds/ion(s) in this sequence contains a specific d-block element, M, then report value of

$$\left(\frac{x+y}{z} \right).$$



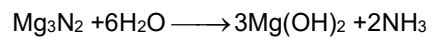
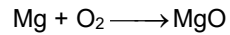
x : Number of various oxidation states of M in species P to T.

y : Number of (O–H) linkages in species P to T.

z : Number of paramagnetic species among P to T.

23. 1.5 M Br_2 solution having unknown volume is reacted with excess of Ferrous thiocyanate solution to form $\text{Fe}_2\text{O}_3 \cdot \text{SO}_4^{2-}$, CN^- along with Br^- . If all the sulphate ions formed are precipitated using BaCl_2 such that 1.864 gm of BaSO_4 is obtained. Calculate volume of Br_2 consumed (in ml).

- 24.** 24 gm pure sample of magnesium is burned in air to form magnesium oxide and magnesium nitride. When products are treated with excess of H_2O , 3.4 gm of gaseous NH_3 is generated according to given reactions.



Calculate the amount of magnesium oxide (in gm) in products.

Answer Key

1.	(D)	2.	(B)	3.	(B)	4.	(C)	5.	(D)	6.	(CD)
7.	(AC)	8.	(CD)	9.	(ABCD)	10.	(BC)	11.	(AC)	12.	(ACD)
13.	(ABC)	14.	(ABC)	15.	(ABD)	16.	(BD)	17.	(AC)	18.	(BCD)
19.	(ABCD)	20.	(B)	21.	(4)	22.	(3)	23.	(17.3)	24.	(28)